Tailoring the Product Selectivity in the Non-Oxidative Methanol Conversion to Dimethyl Ether, Methyl Formate and Dimethoxymethane over a Cu/Hβ Catalyst

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In times of climate change and dwindling fossil fuels, transitioning towards renewable feedstocks is inevitable. Thus, the production of methanol from CO_2 and H_2 is pivotal, since methanol serves as a green base chemical in industry and fuel production. Beyond producing dimethyl ether (DME) *via* methanol dehydration, the non-oxidative conversion of methanol to methyl formate (MF) and dimethoxymethane (DMM) is appealing since H_2 remains a valuable by-product. DME, MF, and DMM are also attractive diesel substitutes or additives.[1] While methanol dehydration to DME requires an acidic catalyst, methanol is transformed to MF over a dehydrogenative catalyst. DMM is formed *via* the dehydrogenation of methanol to formaldehyde, which then acetalizes and condenses with methanol yielding DMM. To perform this reaction in one step, a bifunctional catalyst with both acidic and dehydrogenative sites is required. However, the current state-of-the-art catalysts for these reactions pose challenges, such as low selectivity, activity or stability, accompanied by the thermodynamic limitation of methanol dehydrogenation.[1] Hence, herein, we present the formation of DME, MF and DMM under the same reaction conditions *via* the non-oxidative, continuous gas-phase methanol conversion over a versatile Cu/Hβ catalyst with tailored acidic sites and dehydrogenative properties (Figure 1a).

As shown in Figure 1b, the product spectrum in the continuous gas phase non-oxidative conversion of methanol over Cu/Hβ catalyst can be tailored towards either of the valuable products DME, MF, or DMM with over 75% selectivity under the same reaction conditions (i.e., 240°C, 1 atm, 14549 mL/h/g_{cat}).[2] This is possible by tailoring the bifunctional Cu/Hβ catalyst system according to its acidic and dehydrogenative properties by varying the SiO_2/Al_2O_3 ratio of the Hβ zeolite (25 and 520) and the Cu loading (0.5 and 20 wt.%), respectively. Similar trends were also found in our previous studies.[3,4] Together with catalyst characterizations (XRD, H_2-TPR, CO-DRIFTS, and pyridine FTIR), possible structure-activity relationships are proposed. A high concentration of acidic sites (both Brønsted and Lewis type) of 0.5%Cu/Hβ(25) is

responsible for high DME formation from methanol condensation. When the SiO_2/Al_2O_3 ratio of the Hβ zeolite is increased to 520 by dealumination, only a low Lewis acidic site concentration is maintained, and the acidic sites do not play a dominant role anymore in shaping the product selectivity. Instead, the dehydrogenative properties, linked to the Cu oxidation state, shape the product selectivity towards either MF or DMM over the Cu/Hβ(520) catalysts. The presence of Cu^+ species on 0.5%Cu/Hβ(520) seems to be crucial for high DMM formation and a long induction time of >1500 min suggests a dynamic evolution of an appropriate Cu^+/Cu^0 ratio, being in line with our previous results.[3,4] In contrast, solely Cu^0 species facilitate the formation of MF over 20%Cu/Hβ(520) *via* methanol dehydrogenation, which is in line with literature.[5] Deactivation of both 0.5%Cu/Hβ(520) and 20%Cu/Hβ(520) is observed and linked to changes of the Cu oxidation state, which could successfully be regenerated.

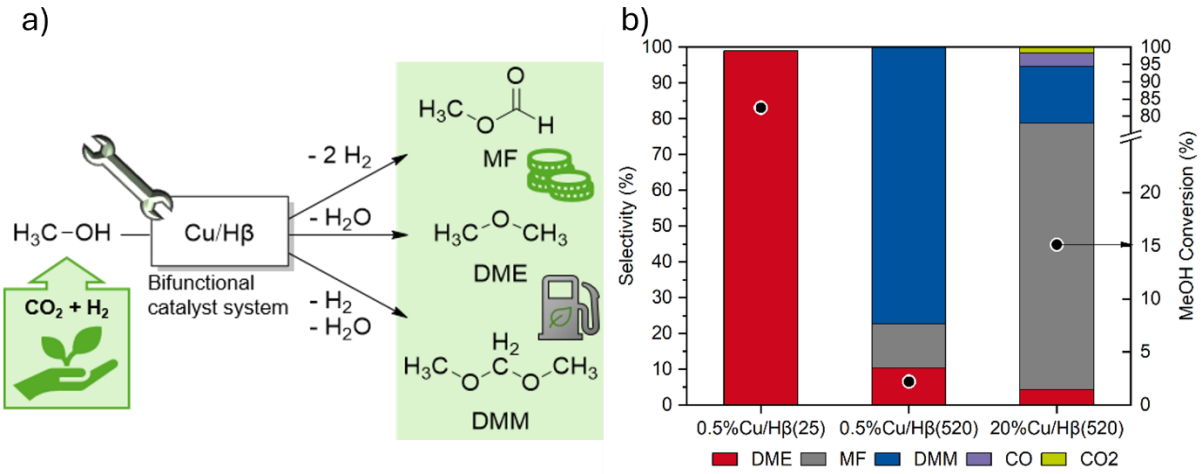

Figure 1. a) Overview of the gas-phase, non-oxidative methanol conversion to the valuable products methyl formate (MF), dimethyl ether (DME), and dimethoxymethane (DMM), which can be obtained under similar reaction conditions by engineering the bifunctional Cu/Hβ catalyst system. b) Screening of Cu/Hβ catalysts in the gas-phase, non-oxidative conversion of methanol. Reaction conditions: 240 °C, 1 atm, GHSV = 14549 mL/h*$g_{cat.}$, 0.1 g of catalyst diluted with 0.9 g of SiC, $n(CH_3OH)/n(N_2)$ = 0.24, N_2 flow rate = 19.4 mL/min, in situ reduction prior to reaction (450 °C, H_2-flow = 20 mL/min, 3 h). Results shown after 1500 min TOS.

Acknowledgments
This work was funded by the German Federal Ministry of Education and Research within the project NAMOSYN: Nachhaltige Mobilität durch synthetische Kraftstoffe (FKZ 03SF0566P0) and by German Research Foundation (PA 1689/22-1). NS acknowledges funding by the German Federal Environmental Foundation (Deutsche Bundesstiftung Umwelt DBU).

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